Mathematical destination

Remove finitely many vertices and every road incident to them. A **forever piece** is a connected component that reaches arbitrarily far, not merely a piece touching an arrow in the drawing. The model is an infinite graph; each page is a finite window with a specified continuation.

The two-way line and the two-rail endless ladder have **two ends**: at most two forever pieces can remain after any finite deletion, and two can be attained. The full square grid has **one end**: every finite deletion leaves exactly one forever piece, though finite pockets may be trapped. The endlessly branching tree has **infinitely many ends**: deleting the center and all vertices within r roads leaves $3 \cdot 2^r$ forever pieces. In particular the counts for $r = 0, 1, 2, 3$ are **3, 6, 12, 24**.

Why these facts hold. On the line and ladder, all blockers lie between two far-out columns, leaving two connected tails. Every forever piece must contain at least one tail. On the grid, all blockers fit inside a box, and the exterior connects around it. In the tree, each branch continues and splits into two without ever rejoining. The detailed arguments below justify the conclusions about infinity; finite drawings alone do not.

What children discover. A barrier can create finite pockets without separating ways to escape forever. Counting arrows is unreliable: several arrows may be joined outside the window. Children experiment and conjecture first, then explain what remains true however far the pattern is extended. An end records a persistent direction of escape as finite regions are removed; it is not the count of components for one arbitrary placement. The tree investigation shows there is no finite bound, without claiming to enumerate its individual ends.

Prerequisites and band map. Grades 3–5 enter by placing blockers and tracing routes. Children need to recognize connected pieces and distinguish "large but finite" from "keeps going." Adults may read and scribe. The box argument and arbitrary-radius tree rule are readiness-dependent older-table continuations. No classroom-tested age fit is claimed.

A K–1 table. There is no K–1 packet. An adult reads Problems 1 and 7 aloud and works them on one shared copy of pages 1 and 4: children place the blockers and run a finger along each road that is left. Ask "Which pieces can you keep walking along forever? Which are shut in?" Key in plain words: on the line, however the blockers sit, two pieces go on forever, one each way, and anything between two blockers is shut in. On the branching map, covering the middle dot leaves 3 pieces that go on forever; covering its 3 neighbours too leaves 6; covering the next 6 dots too leaves 12, with 10 blockers in all. Another week's K–1 packet is the alternative. This route is a prediction; it has not been tried.

Preparation and common launch

Per pair: **ten opaque blockers 6–8 mm across**, pencil, eraser, and four student pages. The smallest grid spacing is 12 mm at 100% print size. Rehearse placing a blocker on one vertex without obscuring a neighboring one; verbally close every incident road. Print single-sided. Prepare four sets: three for two third-grade pairs and an older pair with a rotating referee, each table anchored by an adult, and a fourth, with one copy each of pages 1 and 4, for the K–1 table. Ten blockers cover the tree through radius 2; radius 3 can be reasoned about or marked with Xs.

Let children place and remove blockers. Use the printed four-dot example: cover the upper-right vertex, identify its two closed roads, and trace the two surviving roads as one connected piece. Then, in one minute at the board, draw a ray (a row of dots with an arrow at one end only) and cover its third dot: the first two dots are shut in, a finite trapped piece, and the rest is one forever piece. A ray, not a two-way line, leaves Problem 1 to the children. Show the keep-mark: when a placement is worth keeping, mark X on its dots before lifting the blockers. One partner blocks, the other traces; then swap. Explain the continuation arrows before asking for forever pieces.

First route. Third graders: about 10 minutes on the line (Problems 1–2), then the ladder's Problem 3, then the branching map's Problem 7 with ten blockers and its three counts. Older pairs add Problems 4–6 before the tree as time allows. Close by asking which map gave the most forever pieces. Problem 8 and whatever was skipped make a return visit; finishing all eight problems is not an hour-long requirement. These are planning estimates.

Numbered answers and hints

Problem 1. The maximum is **2** with one, two, or four blockers. Every placement on distinct line vertices leaves the left and right tails. Any surviving pieces between the first and last blockers are finite. Adjacent blockers enclose no open piece. *Hint:* extend the same picture several steps farther; which pieces can keep growing?

Problem 2. Number the nine visible dots 1–9 from left to right. Block **2, 4, 5, 7**. Dots **3 and 6** are exactly the two trapped pieces; the left and right tails still count as two forever pieces. Other placements work. Do not count a covered dot or the empty space between adjacent blockers as a component.

Answers continued and the reasons they hold

Problem 3. The fewest is **two**. One blocked ladder dot leaves the graph **connected**: use the other rail to bypass it. Two blockers make **two forever pieces** when they cover both dots of one rung, or a **diagonal pair**: one rail at a column and the other rail at the next column. No road then crosses between those two columns. On the printed ladder 25 two-dot placements split it: 9 rungs and 16 diagonal pairs. For a two-dot placement that does not split it, cover two dots on the same rail, or one on each rail two or more columns apart; the other rail and surviving rungs still join every open dot. *Hint:* make an actual detour before deciding that a gap in one rail is a separation; a diagonal pair passes that test. *Follow-up:* "Can you split it with two dots that are not on the same rung?"

Problem 4. No. All four blockers lie between some leftmost and rightmost column. Beyond each side is one intact, connected ladder tail. The intervening ladder has finitely many vertices, so every forever piece reaches a tail. Each tail belongs to one component, giving at most two forever pieces.

Four blockers can make *three components*, but that is a different count: block both vertices at each of two rungs with one intact rung between. The middle rung is a finite two-dot pocket; the two outside pieces are infinite. *Hint:* where could a third forever piece go after passing all blocked columns? Far-away rungs are still present.

Problem 5. On the left, leave a dot open (choose a dot not on the window's edge) and block its four immediate neighbors: above, below, left, right. This isolates that dot. All other open dots belong to **one forever piece**; the isolated dot does not count. On the right, **one forever piece** contains every open dot, including P and Q.

A concrete right-grid route is **4 up, 4 right, 4 down**, going around the top of the seven-X wall, beyond the printed window. It has **12 steps**; the reflected bottom route also works. Although shortest is not requested, 12 is optimal: a route must cross the wall's column at least four rows above or below P, requiring eight vertical steps and at least four horizontal steps. Every route leaves the printed window. *Hint:* a finite wall stops somewhere; the dots past the arrows are open, so trace around its top or bottom instead of treating the edge of the page as a wall.

Problem 6. No finite collection can produce two forever pieces. Enclose every blocker in a square of grid vertices. Its exterior is connected: given two exterior points, choose a still larger square containing both. From each point, move outward along a coordinate already outside the blocked box to this larger boundary, then join them along the boundary. Every infinite component contains a point in this exterior, since the box has only finitely many vertices. All such points are connected, so there is exactly one infinite component. Finite pockets inside the box are allowed.

Hint sequence: extend the barrier past the paper; ask whether all its *finitely many* blockers fit in a box; then offer the exterior route. Unsuccessful experiments alone cannot prove impossibility.

Problem 7. Covering only the center leaves **3** forever pieces. Covering the center and its three neighbors leaves **6**. Covering through two roads leaves **12**. These stages use **1, 4, and 10 blockers** in total. Count surviving branches, not removed vertices. Each branch is infinite by the stated continuation rule, and different branches cannot reconnect because the model has no cycles. *Hint:* follow one of the three initial branches and see what happens when its first remaining vertex is removed.

Problem 8. Blocking through three roads leaves **24** forever pieces and would cover **22 vertices**. Each further radius doubles the count: each outgoing branch is replaced by its two child branches. Starting from 3 proves $3 \cdot 2^r$

by repeated doubling. This produces arbitrarily many separate forever pieces using a finite blocked ball, hence infinitely many ends. It does not mean the tree has only 24 ends, or that all ends are counted by the arrows currently visible.

Sources and checking limits

Druțu and Kapovich, *Geometric Group Theory*, §9.1, especially Definition 9.3 and Exercise 9.4, printed pp. 287–289 (PDF 307–309), supplies the ends and unbounded-component framework. The blockers, small maps, and prompts are authored specializations.

The independent standard-library checker verifies the line example, contrasting ladder deletions, grid pocket and 12-step wall route, and branching counts. Its finite windows do *not* prove assertions about infinity; the tail, exterior, and continuing-branch arguments do. Final student pages and every guide page were digitally inspected. This remains an **unpiloted prototype with no physical setup rehearsal**; fit and pacing need testing.